

# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE

### Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **geotechnical analysis** for retaining structures and foundations on large-scale European infrastructure projects, contributing to **Metrolink** and **Fehmarnsund Link** designs.
- Prepared detailed **geotechnical design reports**, technical specifications, and conducted site walkovers for pharmaceutical and commercial developments, ensuring compliance with project requirements.
- Managed and interpreted **geotechnical ground investigations**, providing on-site technical support and supervision during the construction phase for major civil engineering schemes.
- Developed and applied **numerical analysis** techniques for complex soil-structure interaction problems, enhancing the accuracy of earthworks and slope stability assessments.
- Contributed to the integration of **digital tools and automation** within project delivery workflows, improving efficiency in data processing and design iterations.
- Mentored and provided technical guidance to junior engineers on **geotechnical design principles** and project execution, fostering a collaborative team environment.
- Containerized simulation tools using **Docker** and deployed them on **AWS EC2** instances for on-demand analysis, reducing setup time for team members by approximately **40%**.

## SKILLS

**Geotechnical Engineering** – Foundations, Retaining Structures, Slope Stability, Earthworks, Soil Mechanics, Ground Investigations, Site Supervision

**Software & Tools** – GIS, Numerical Analysis Software, AutoCAD, Civil 3D, Python (for scripting/automation)

**Analysis & Design** – Finite Element Analysis, Geotechnical Design, Structural Analysis, Risk Assessment

**Project Management** – Technical Specifications, Design Reports, Site Walkovers, Client Communication, Stakeholder Management

**Cloud & DevOps** – AWS (EC2, S3), Docker, Jenkins, Git

## PROJECTS

### Geotechnical Design for Urban Rail Expansion

- Developed detailed **geotechnical designs** for deep excavations and shallow foundations supporting a major urban rail expansion project, adhering to stringent safety and performance standards.
- Performed **slope stability analysis** using advanced numerical modelling techniques, simulating various ground conditions and loading scenarios to ensure long-term structural integrity.
- Collaborated with civil engineers and project managers to integrate geotechnical findings into the overall project plan, facilitating timely UAT and construction readiness.

### Automated Data Processing for Site Investigations

- Created **Python scripts** to automate the parsing and initial analysis of borehole data and CPT logs from ground investigations, reducing manual processing time by **~30%**.
- Developed data validation routines to identify anomalies and inconsistencies in geotechnical survey results, improving the reliability of input data for design models.
- Integrated the data processing workflow with **AWS S3** for secure storage and version control of raw and processed investigation data.

## EDUCATION

**M.Sc. Data Analytics** — National College of Ireland, Dublin

Feb 2026

**B.E. Computer Science** — SNS College of Technology, Coimbatore, India

2021

## CERTIFICATIONS

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- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)